

EXP NO: 8

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Cases are created, and two test cases are written for at least five user stories (happy path and error scenarios) using Azure DevOps.

Test Planning and Test Case Design Procedure

1. Understand Core Features of the Application

- User Registration & Login
- Student Profile Management
- Course Enrollment
- Attendance Management
- Marks & Report Generation

2. Define User Interactions

Each test case represents real user actions

Example:

- Student logs in
- Updates profile
- Enrolls in a course
- Views attendance
- Downloads report

3. Design Happy Path Test Cases

- Verify system works under normal conditions
- Example: User logs in successfully, Student enrolls in course, Report downloaded successfully

4. Design Error Path Test Cases

- Test invalid or unexpected scenarios
- Example: Login with wrong password, Enrollment when seats are full, Invalid data in profile update

5. Break Down Steps and Expected Results

- Each test case includes:
 - Step-by-step actions
 - Expected output

6. Use Clear Naming and IDs

Test cases are named clearly using IDs such as:

- TC01 – Successful Login
- TC02 – Login Failure
- TC05 – Course Enrollment Success
- TC10 – Report Download Failure

This helps in:

- Easy identification of test cases
- Linking test cases with user stories and features
- Better tracking in **Azure DevOps**

7. Separate Test Suites

Test cases are grouped based on functionality:

- TS01 – User Login
- TS02 – Student Profile
- TS03 – Course Enrollment
- TS04 – Attendance Management
- TS05 – Reports & Results

This improves:

- Organization of testing
- Easy navigation
- Efficient test execution in Azure DevOps

8. Prioritize and Review

Critical user actions are marked as High Priority, such as:

- Login functionality
- Course enrollment
- Attendance marking
- Report generation

Each test case is:

- Reviewed for completeness
- Verified against user story requirements
- Ensured for proper coverage of both happy and error scenarios

This ensures:

- Better quality assurance
- Accurate validation of system functionality

1.New test plan

Azure DevOps

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Student Data Manage... +

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New Test Plan

Name *
Student Data Management System Test Plan

Area Path *
Student Data Management System

Iteration *
Student Data Management System\Sprint 1 2/3/2026 - 2/7/2026

Create Cancel

2. Test suite

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Student Data Man...
Feb 3 - Feb 7
100% run, 100% passed
Past
View report

Test Suites
Filter suites by name
Student Data Management System Te...

- New Suite
 - Static suite
 - Requirement based suite
 - Query based suite
- Assign configurations
- Export
- Assign testers to run all tests
- Import test suites

TS03 – Course Enrollment (ID: 114)
Define Execute Chart
Add a test case
Use this tab to collate, add and manage test cases
New Test Case

3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Student Data Management System – Test Plans

USER STORIES

- As a user, I want to log in using email and password so that I can access the system. (ID: 28).
- As a student, I want to view my personal and academic details so that I can verify my records. (ID: 14).
- As a student, I want to enroll in courses so that I can attend classes. (ID: 31).
- As a student, I want to view my attendance so that I can track my eligibility. (ID: 20).
- As a student, I want to download my report card so that I can use it officially. (ID: 30).

Test Suites

Test Suit: TS01 - User Login (ID: 112)

1. TC01 – Successful Login

- **Action:**
 - Go to Login page
 - Enter valid email and password
 - Click “Login”
- **Expected Results:**
 - User is logged in and redirected to dashboard
- **Type:** Happy Path

2. TC02 – Login Failure

- **Action:**
 - Go to Login page.
 - Enter valid email and wrong password.
 - Click "Login".
- **Expected Results:**
 - Error message “Invalid credentials” is displayed
- **Type:** Error Path

Test Suit: TS02 – Student Profile (ID: 113)

1. TC03 – View Profile

- **Action:**
 - Login successfully.
 - Navigate to Profile page.
- **Expected Results:**
 - Student details are displayed correctly
- **Type:** Happy Path

2. TC04 – Invalid Profile Update

- **Action:**
 - Go to Profile Page
 - Enter Invalid Phone/Email
 - Click “Save”
- **Expected Results:**
 - Error message "Invalid input" is displayed.
- **Type:** Error Path

Test Suit: TS03 – Course Enrollment (ID: 114)

1. TC05 – Successful Course Enrollment

- **Action:**
 - Login
 - Select course
 - Click “Enroll”
- **Expected Results:**
 - Enrollment is successful
- **Type:** Happy Path

2. TC06 – Enrollment Failure

- **Action:**
 - Login
 - Select a full course
 - Click “Enroll”
- **Expected Results:**
 - Error message “Course is full” is displayed
- **Type:** Error Path

Test Suit: TS04 - Attendance (ID: 115)

1. TC07 – Mark Attendance Successfully

- **Action:**
 - Faculty login
 - Select class
 - Mark Attendance
 - Click “Save”
- **Expected Results:**
 - Attendance is recorded successfully
- **Type:** Happy Path

2. TC08 – Attendance Save Failure

- **Action:**
 - Mark attendance
 - Simulate network/server error
 - Click "Save".
- **Expected Results:**
 - Error message "Failed to save attendance" is displayed.
- **Type:** Error Path

Test Suit: TS05 - Reports (ID: 116)

1. TC09 – Download Report Card

- **Action:**
 - Login
 - Navigate to Reports
 - Click “Download Report”
- **Expected Results:**
 - Report is downloaded successfully
- **Type:** Happy Path

2. TC10 – Report Download Failure

- **Action:**
 - Click "Download Report".
 - Simulate server issue.
- **Expected Results:**
 - Error message: "Unable to download report" is displayed
- **Type:** Error Path

Test Cases

TEST CASE 117

117 TC01 – Successful Login

Karthik Sah E

0 Comments Add Tag

Save and Close Follow

Updated by Karthik Sah E: 23m ago

Status Design Area Student Data Management System Reason New Iteration Student Data Management System/Sprint 1

Steps

Steps Action Expected result Attachments

1. Enter valid email User logged in successfully

2. Enter password User logged in successfully

3. Click login User logged in successfully

Click or type here to add a step

Recent test results

Passed by Karthik Sah E with Windows 10 Completed 7:09 AM, 4/6/2026

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Add an existing work item as a parent

Status

Priority 2

Automation status Not Automated

Parameter values

TEST CASE 118

118 TC02 – Login Failure

Karthik Sah E

0 Comments Add Tag

Save and Close Follow

Updated by Karthik Sah E: 12m ago

Status Design Area Student Data Management System Reason New Iteration Student Data Management System/Sprint 1

Steps

Steps Action Expected result Attachments

1. Enter valid email Error message displayed

2. Enter wrong password Error message displayed

3. Click login Error message displayed

Click or type here to add a step

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

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Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

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Status

Priority 2

Automation status Not Automated

Parameter values

4. Installation of test

chrome web store

Search extensions and themes

Discover **Extensions** Themes

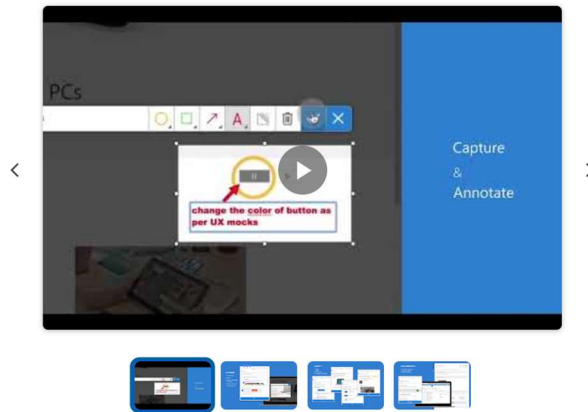


Test & Feedback

Add to Chrome

4.2 ★ (176 ratings) ⓘ [Share](#)

Extension Workflow & Planning 200,000 users



chrome web store

Search extensions and themes

Discover **Extensions** Themes

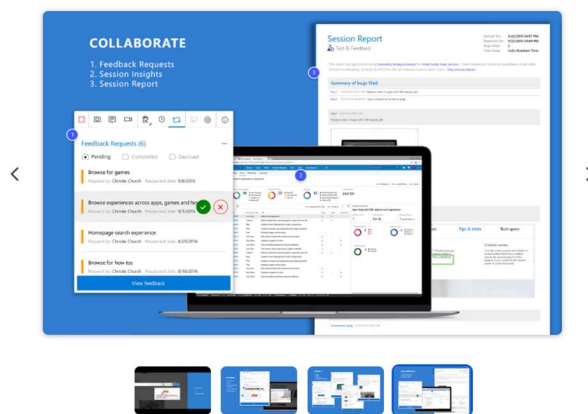


Test & Feedback

Remove from Chrome

4.2 ★ (176 ratings) ⓘ [Share](#)

Extension Workflow & Planning 200,000 users



5. Test and feedback Showing it as an extension

The screenshot displays the Azure DevOps Test Plans interface. The left sidebar shows the navigation menu with 'Test Plans' selected. The main area shows the test suite 'TS01 - User Login (ID: 112)' with a status of '100% run, 100% passed'. The test points are listed as follows:

Test Points (2 items)
☐ Title
☐ TC01 - Successful Login
☐ TC02 - Login Failure

An 'Extensions' popup is visible, showing the following extensions:

Extension	Configuration	Current Tester
McAfee WebAdvisor	Windows 10	Karthik Sah E
Test & Feedback	Windows 10	Karthik Sah E

6. Running the test cases

Student Data Manage...

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Student Data Man...

Feb 3 - Feb 7 **Past**

100% run, 100% passed. [View report](#)

Test Suites

Filter suites by name

Student Data Management System Te...

TS05 - Reports

TS04 - Attendance

TS03 - Course Enrollment

TS02 - Student Profile (2)

TS01 - User Login (2)

TS01 - User Login (ID: 112)

Define Execute Chart

Test Points (2 items)

Title	Outcome	Order	Test Case Id	State	Configuration	Current Tester
TC01 - Successful Login	Passed	1	117	Design	Windows 10	Karthik Sah E
TC02 - Login Failure	Passed	2	118	Design	Windows 10	Karthik Sah E

View execution history

Mark Outcome

Run

Reset test to active

Edit test case

Assign tester

View test result

Run for web application

Run for desktop application

Run with options

Save and close | Create bug

117*: TC01 - Successful Login

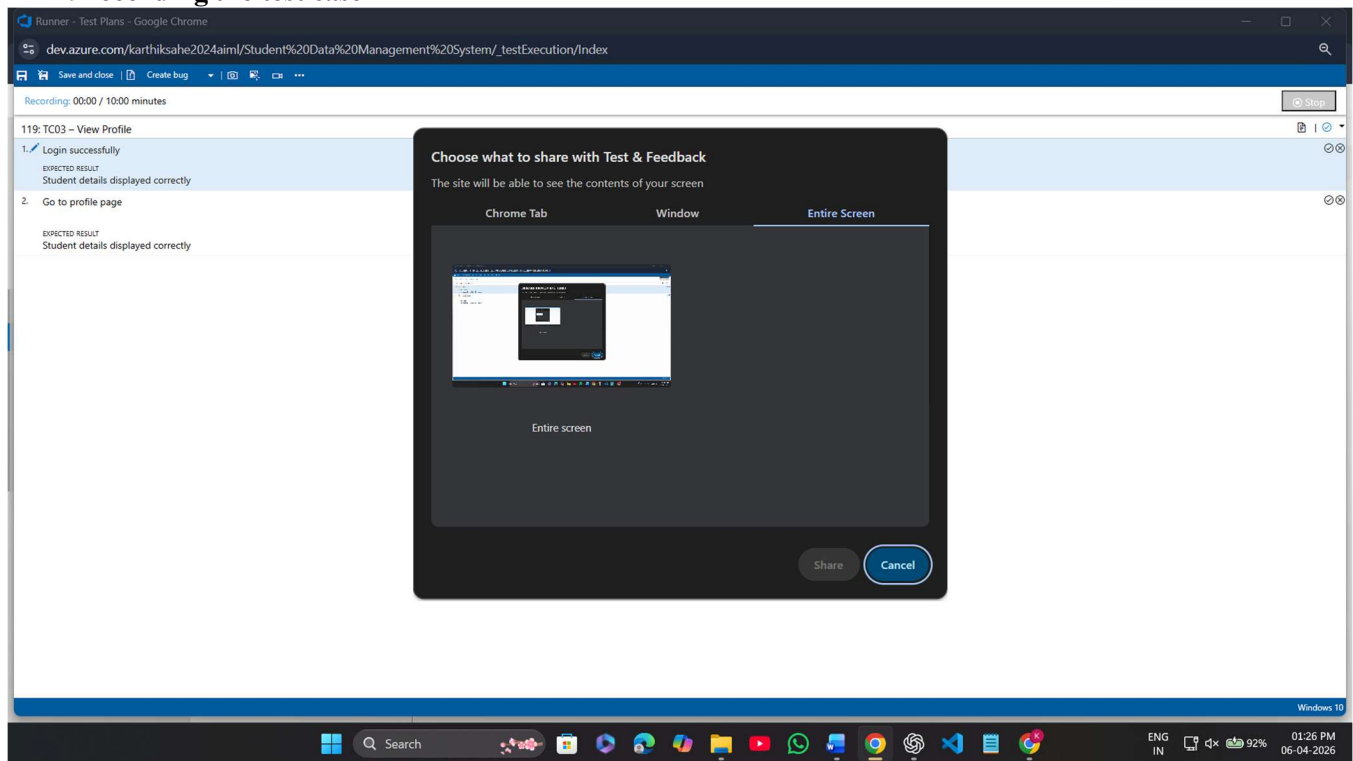
1. Enter valid email
EXPECTED RESULT
User logged in successfully

2. Enter password
EXPECTED RESULT
User logged in successfully

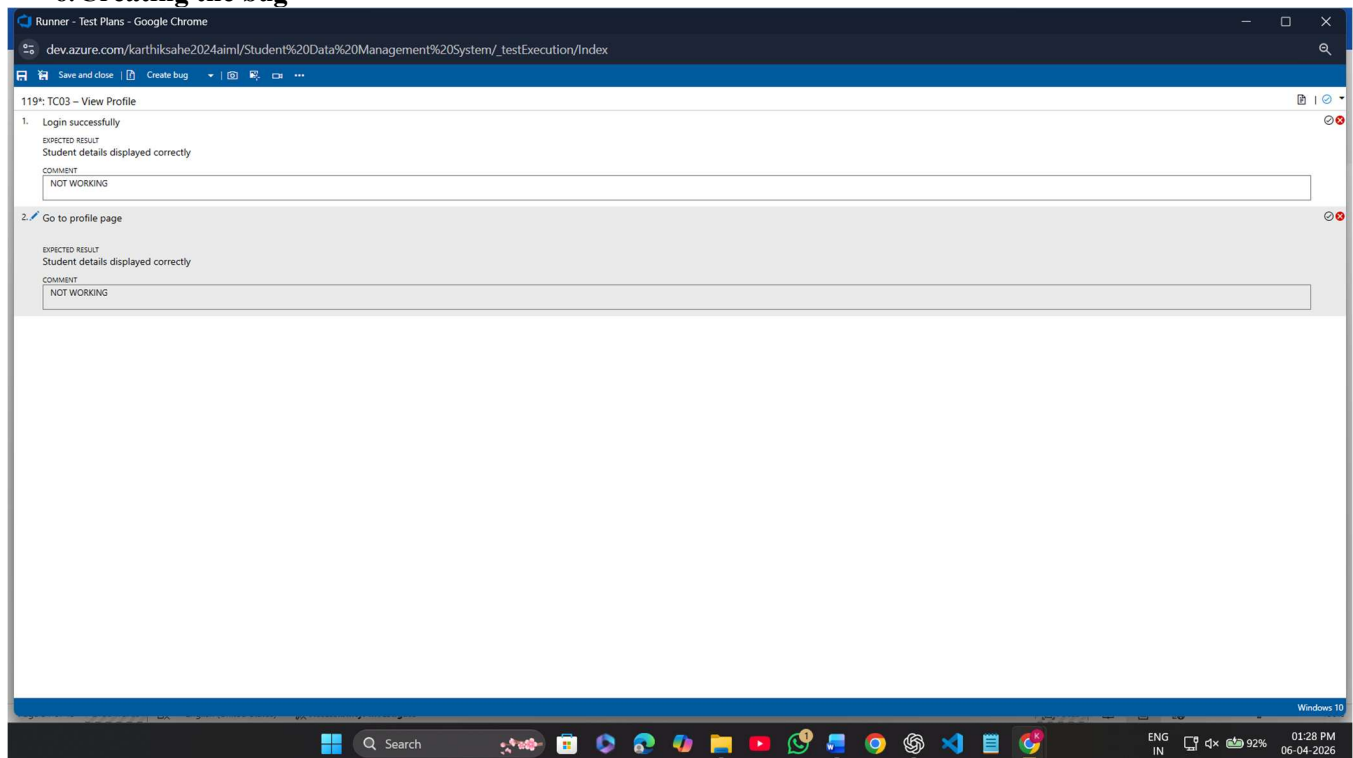
3. Click login
EXPECTED RESULT
User logged in successfully

Windows 10

7. Recording the test case



8. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/karthiksahe2024aiml/Student%20Data%20Management%20System/_testExecution/Index

NEW BUG *

Profile Update Accepts Invalid Data

No one selected0 CommentsAdd Tag

Save and Close

State: NewReason: NewArea: Student Data Management SystemIteration: Student Data Management System\Sprint 1

Details

Repro Steps

4/6/2026 7:58 AM Bug filed on "TC03 - View Profile"

Step no.	Result	Title
1.	Failed	Login successfully
Expected Result		
Student details displayed correctly		
Comments: NOT WORKING		
2.	Failed	Go to profile page
Expected Result		
Student details displayed correctly		
Comments: NOT WORKING		

Test Configuration: Windows 10

System Info

Browser - Name	Google Chrome 146
Browser - Language	en-US
Browser - Height	816
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64

Planning

Resolved Reason

Story Points

Priority: 2

Severity: 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Add an existing work item as a parent

Tested By

119 TC03 - View Profile

Updated 39m ago

Design

System Info

Found in Build

Integrated in Build

Runner - Test Plans - Google Chrome

dev.azure.com/karthiksahe2024aiml/Student%20Data%20Management%20System/_testExecution/Index

NEW BUG *

Profile Update Accepts Invalid Data

No one selected0 CommentsAdd Tag

Save and Close

State: NewReason: NewArea: Student Data Management SystemIteration: Student Data Management System\Sprint 1

Details

Repro Steps

System Info

Browser - Name	Google Chrome 146
Browser - Language	en-US
Browser - Height	816
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64
Operating system - Processor model	12th Gen Intel(R) Core(TM) i5-12450H
Operating system - Number of processors	12
Memory - Available	5548032000
Memory - Capacity	16802910208
Display - Pixels per inch (X axis)	120
Display - Pixels per inch (Y axis)	120
Display - Device pixel ratio	1.25

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Resolved Reason

Story Points

Priority: 2

Severity: 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

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Related Work

Add link

Add an existing work item as a parent

Tested By

119 TC03 - View Profile

Updated 40m ago

Design

System Info

Found in Build

Integrated in Build

9. Test case results

Azure DevOps

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Filter on

119 TC03 - View Profile

Windows 10

119 TC03 - View Profile

Configuration Windows 10

Summary

Attachments 2

Summary

Run by Karthik Sah E

Pipeline run tested None

Configuration Windows 10

Completed time 4/6/2026 8:03 AM

Test plan 108 Student Data Management System Test Plan

Test suite 113 TS02 - Student Profile

Test case 119 TC03 - View Profile

Priority 2

Analysis

Analysis owner Karthik Sah E

Failure type None

Resolution None

Comment

Linked work items 1

+ Add

ID ↓	Work item title	State	Assigned To	Created date	Updated date
121	Profile Update Accepts Invalid Data	New	Unassigned	22m ago	22m ago

Test failed

Start time 4/6/2026 7:56 AM

Duration 6m 46s

Show images

Attachments

SystemInformation-2026-04-06T07-57-20.67672.json (534 B)

SystemInformation-2026-04-06T07-58-55.6207.json (534 B)

Step	Outcome	Action	Expected Result
1	Failed	Login successfully CommentNOT WORKING	Student details displayed correctly
2	Failed	Go to profile page CommentNOT WORKING	Student details displayed correctly

10. Test report summary

BUG 121

121 Profile Update Accepts Invalid Data

Karthik Sah E

0 Comments Add Tag

Save and Close

Follow

Updated by Karthik Sah E Just now

Details

Repro Steps

4/6/2026 7:58 AM Bug filed on "TC03 - View Profile"

Step no.	Result	Title
1.	Failed	Login successfully
		Expected Result Student details displayed correctly
		Comments: NOT WORKING
2.	Failed	Go to profile page
		Expected Result Student details displayed correctly
		Comments: NOT WORKING

Test Configuration: Windows 10

System Info

Browser - Name	Google Chrome 146
Browser - Language	en-US
Browser - Height	816
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64
Operating system - Processor model	12th Gen Intel(R) Core(TM) i5-12450H
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Planning

Resolved Reason

Story Points

Priority 2

Severity 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

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Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Add an existing work item as a parent

Tested By

119 TC03 - View Profile

Updated 1h ago @ Design

System Info

Found in Build

Integrated in Build

BUG 121

121 Profile Update Accepts Invalid Data

Karthik Sah

0 Comments Add Tag

Save and Close

Follow

Updated by Karthik Sah E Just now

State New

Area Student Data Management System

Reason New

Iteration Student Data Management System(Sprint 1)

Repro Steps

4/6/2026 7:58 AM Bug filed on "TC03 - View Profile"

Step no.	Result	Title
1.	Failed	Login successfully
		Expected Result
		Student details displayed correctly
		Comments: NOT WORKING
2.	Failed	Go to profile page
		Expected Result
		Student details displayed correctly
		Comments: NOT WORKING

Test Configuration: Windows 10

System Info

Browser - Name	Google Chrome 146
Browser - Language	en-US
Browser - Height	816
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x86_64
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Resolved Reason

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Completed

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Related Work

Add link

Add an existing work item as a parent

Tested By

119 TC03 - View Profile

Updated 1h ago @ Design

System Info

Found in Build

11. Progress report

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Student Data Management System Test Plan

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Outcome

Configuration

Tester

Priority

Assigned To

Summary

1 Test plans

10 Test points

10 (10 / 10) Test points run

100% Run

100% (10 / 10) Pass rate

10 Passed

Outcome trend

Last 14 Days

0 2 4 6 8 10

2026-03-31 2026-04-01 2026-04-02 2026-04-03 2026-04-04 2026-04-05 2026-04-06 2026-04-07 2026-04-08 2026-04-09 2026-04-10 2026-04-11 2026-04-12 2026-04-13 2026-04-14

Not run Passed

Details

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
Student Data Management System Test Plan	10	100	100	0	0
TS01 - User Login	2	100	100	0	0
TS02 - Student Profile	2	100	100	0	0
TS03 - Course Enrollment	2	100	100	0	0
TS04 - Attendance	2	100	100	0	0
TS05 - Reports	2	100	100	0	0

12. Changing the test template

Azure DevOps karthiksah2024aiml / Settings / Process

Search

Organization Settings
karthiksah2024aiml

Search Settings

General

- Overview
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- Users
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- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools
- Parallel jobs
- OAuth configurations

Repos

- Repositories

Artifacts

- Storage

All processes

Processes Fields

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile (default)	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	1
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of decisions.	0

Help Filter by process name

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path.